

C1 · Parsing-Retrieval Disconnect

Candidate parsers



Cleaner-looking ≠ better retrieval

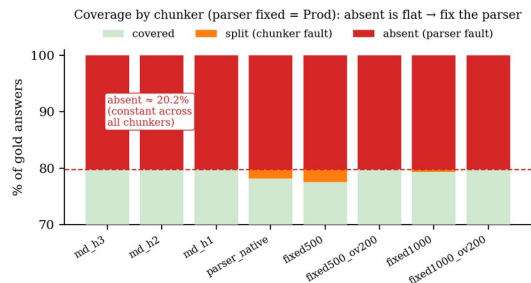
parser	Boundary Clarity	answer	retrieval
MinerU	0.72 (highest)	no	MISS
Prod (ours)	lower	yes	HIT

highest Boundary Clarity, yet worst retrieval:
appearance metrics measure formatting, not content

anti-correlation $r = -0.81$

C2 / C3 · Diagnose + Select

C2 · Coverage diagnostic (no retriever)



20.2% of answers absent = parser fault
flat across 8 chunkers → fix the parser, not the chunker

C3 · RCPS selection protocol (no training)

held-out Q-A probe, scored over 3 retrievers:



BGE-M3 · mE5 · Qwen3-Emb

→ rank parsers & chunkers by retrieval

C4 · Bounded Parser Training

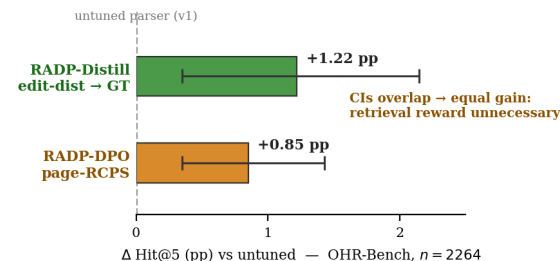
C4 · Parser-side training

best-of-K parses from Prod, ranked two ways:

RADP-Distill — edit-distance → GT

RADP-DPO — page-local RCPS

Result



the two signals tie → retrieval reward unnecessary
the deployable lever is fidelity distillation